

Problem Set 5 — Uniform Acceleration

Physics 2026–27 · due at the start of class, Monday, September 28

NAME _____ DATE _____

How to work this set. Five variables — v_i , v_f , a , t , Δx — four equations, one selection rule: *find the one variable you neither know nor want, and use the equation that omits it*. Name that missing variable in your E row every time. Symbols first, numbers last, units the whole way.

1 Pick the equation — don't solve

SELECTION RULE

For each scenario, name the **missing variable** (neither known nor wanted) and write the equation you'd use. No arithmetic.

- a. A jet starts from rest and accelerates at 3.5 m/s^2 down 900 m of runway. You want its takeoff speed.
- b. A train slows from 40 m/s to 10 m/s over 60 s . You want how far it traveled.
- c. A dropped phone falls for 0.80 s . You want how far it fell.
- d. A sprinter reaches 9.0 m/s in 3.0 s from rest. You want her acceleration.

ANSWERS

2 The rocket sled

KDEA

A rocket sled starts from rest and accelerates at a constant 12 m/s^2 for 5.0 s . Two targets, two KDEA passes:

- a. Its final velocity.
- b. The distance it covers.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 No time given — on purpose

KDEA

A car moving at 22 m/s brakes with constant acceleration and stops in 40 m . Find the acceleration (sign included). Notice: the word "time" appears nowhere in this problem — let that steer your E row.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 No acceleration given

KDEA

A runner speeds up steadily from 2.0 m/s to 8.0 m/s over 3.0 s . How far does she travel during those 3.0 s ? (Acceleration is constant, but its value never appears — you don't need it.)

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5 Two-stage trip

MULTI-STEP

A motorcycle starts from rest, accelerates at 4.0 m/s^2 for 6.0 s , then holds a constant velocity for another 8.0 s .

- How fast is it moving at the end of the first stage?
- How far does it travel during the first stage?
- How far during the second stage? (Careful — constant velocity means the four-equation bank isn't even needed here.)
- Total distance for the whole trip?

WORK

6 Speeding up or slowing down?

SIGNS

For each, state whether the object is speeding up or slowing down, using the sign rule (same signs vs. opposite signs):

a. $v = +14 \text{ m/s}$, $a = -3 \text{ m/s}^2$

b. $v = -6 \text{ m/s}$, $a = -2 \text{ m/s}^2$

c. $v = -9 \text{ m/s}$, $a = +4 \text{ m/s}^2$

d. One sentence: why does "negative acceleration" alone tell you nothing about slowing down?

7 Grade this work

CRITIQUE

The problem: "A car slows uniformly from 20 m/s to rest in 5.0 s. How far does it travel while stopping?" A student writes:

$$\Delta x = v_i t + \frac{1}{2} a t^2 = (0)(5.0) + \frac{1}{2}(-4.0)(25) = -50 \text{ m}$$

- a. The equation choice is fine and $a = -4.0 \text{ m/s}^2$ is correct. Find the actual error. (Hint: read the subscripts out loud, Workshop-I style.)
- b. Fix it: run the full KDEA and report the real stopping distance. Sanity-check the sign of your answer.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Graded on the homework 4-3-2-1 scale. Every E row must name the missing variable — that one habit is most of this week.